

Diagnostic

Name _____ Course BioE 147 / 247 (circle one)

This is not graded and does not enter your grade in any way. Its only purpose is to tell me where this room actually is, so I can set the pace honestly rather than guess.

You already told me what you have *used*, in the form you filled in before today. These four questions are the other half: what you can *do* right now, cold, with nothing to look at.

They come from four different backgrounds on purpose. **Almost nobody answers all four.** If you have never seen something, leave it blank — a blank is more useful to me than a guess, and it costs you nothing. Do not look anything up; a looked-up answer makes me teach the wrong course to the wrong room.

1. Molecular

A gene is preceded by a promoter. A protein called a *repressor* binds a site that overlaps that promoter.

In one or two sentences: what physically stops the gene being expressed, and what would happen to expression if you doubled the number of repressor molecules in the cell?

2. Dynamical

A protein is made at a constant rate α (molecules per minute) and each molecule is removed with probability per unit time γ , so

$$\frac{dx}{dt} = \alpha - \gamma x.$$

(a) What is the steady-state value of x ?

(b) Starting from $x = 0$, roughly how long does it take to get most of the way there? Give the answer in terms of α and γ .

3. Physical

An *E. coli* cell is about **1 femtolitre** (10^{-15} L).

Approximately how many molecules are in that cell at a concentration of **1 nM**? (Avogadro's number is $6 \times 10^{23} \text{ mol}^{-1}$.) An order of magnitude is a complete answer; show whatever working you use.

4. Computational

```
import numpy as np

def f(v):
    return np.array([v[1], -v[0]])

x = np.array([1.0, 0.0])
for _ in range(3):
    x = x + 0.1 * f(x)
print(x)
```

What does this print, approximately? And what is this code doing?

Hand this in on your way out. Nothing here is recorded against your name in the gradebook; I read them, I set the pace, and that is the end of it.